DETECTION OF FLYING OBJECT USING HYBRID DEEP LEARNING TECHNIQUES

Project Title and Student Details

Name	Registration Number
Haris K	24BIT0468
Karan Chander S R	24BIT0371

3. Dataset Description and Sample Data

3.1 Dataset sources and description

The study uses a merged, image-based flying-object dataset obtained from AOD-4 and Multi-Background Airborne Object Detection (MBAOD). The source images and annotations were consolidated into a common six-class YOLO format. Unlike a tabular CSV dataset, each record consists of an image and its corresponding object annotations; consequently, the relevant dataset dimensions are image count, class labels and annotated-object count rather than rows and feature columns.

Dataset	Source / DOI	Description
AOD-4	https://doi.org/10.1016/j.dib.2024.110801	22,516 images; four original classes: airplane, bird, drone and helicopter.
MBAOD	https://doi.org/10.17632/zfc8hndczp.1	Six airborne classes and varied backgrounds, object scales and weather conditions.
Unified clean dataset	Local project dataset: comb DSF	57,622 pooled images; 55,409 retained after cleaning and deduplication.

3.2 Dataset characteristics

Attribute	Value
Data type	Image-based, supervised multi-class object detection
Image formats	JPG / JPEG / PNG
Annotation format	YOLO text: class_id, x_center, y_center, width, height (normalized)
Classes	6 — airplane, bird, helicopter, kite, parachute and UAV
Pooled image count	57,622
Clean retained images	55,409
Annotated object instances	86,254 bounding boxes
Negative/background images	1,174 images with no annotated objects
Dataset split recorded in audit	38,786 train; 8,311 validation; 8,312 test
Image resolution	Source samples vary; supplied examples 1024 × 1024, and cGAN composites 640 × 640

The class label is an integer from 0 to 5. Each annotated object is represented by a bounding box. Because a single image can contain more than one object, annotated-object counts must not be interpreted as counts of unique images within each class.

3.3 Class labels and original class distribution

ID	Class	Train boxes	Val boxes	Test boxes	Total boxes	Share
0	Airplane	11,612	2,488	2,488	16,588	19.23%
1	Bird	11,846	2,538	2,538	16,922	19.62%
2	Helicopter	11,445	2,453	2,453	16,351	18.96%
3	Kite	6,650	1,425	1,425	9,500	11.01%
4	Parachute	6,650	1,425	1,425	9,500	11.01%
5	UAV (Drone)	12,167	2,609	2,617	17,393	20.16%
Total	**All classes**	**60,370**	**12,938**	**12,946**	**86,254**	**100%**

The original six-class distribution is imbalanced. Kite and Parachute have 9,500 annotated instances each (approximately 11.01% per class), whereas UAV has 17,393 instances (approximately 20.16%). The shares are calculated using the total 86,254 annotated objects.

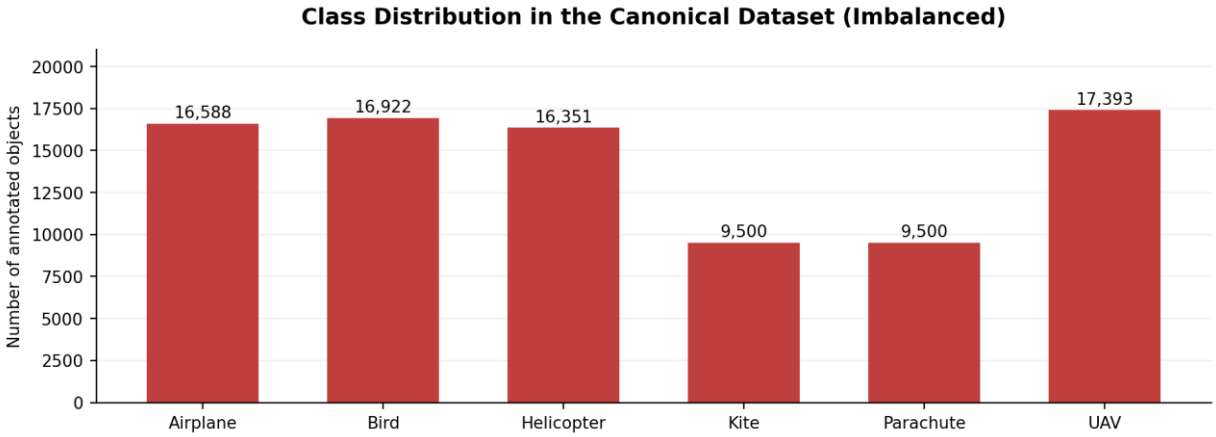


Figure 1. Imbalanced class distribution in the canonical cleaned dataset (annotated objects).

3.4 Two sample images per class

For each of the six categories, two existing source images are shown: a positive example in which the named class is present (YES), and a negative example in which it is absent (NO). The positive bounding box is reproduced from the YOLO label file. The negative sample may contain a different airborne-object class; this illustrates class presence and absence without redefining the task as binary classification.

Sample 1 — YES: Airplane **Sample 2 — NO: Airplane**



Figure 2. Two source-image examples for Airplane: class present (YES) and absent (NO).

Sample 1 — YES: Bird **Sample 2 — NO: Bird**

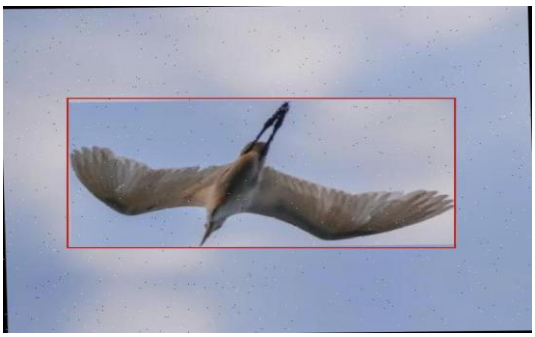




Figure 3. Two source-image examples for Bird: class present (YES) and absent (NO).

Sample 1 — YES: Helicopter



Sample 2 — NO: Helicopter



Figure 4. Two source-image examples for Helicopter: class present (YES) and absent (NO).

Sample 1 — YES: Kite



Sample 2 — NO: Kite



Figure 5. Two source-image examples for Kite: class present (YES) and absent (NO).

Sample 1 — YES: Parachute



Sample 2 — NO: Parachute



Figure 6. Two source-image examples for Parachute: class present (YES) and absent (NO).

Sample 1 — YES: UAV (Drone)

Sample 2 — NO: UAV (Drone)



Figure 7. Two source-image examples for UAV (Drone): class present (YES) and absent (NO).

3.5 Data imbalance and recorded preprocessing comparison

Dataset cleaning removed 2,210 duplicate or conflicting source images and three invalid images, reducing the image count from 57,622 to 55,409. This is a data-cleaning operation and is not equivalent to class balancing.

Processing stage	Image count	Change
Original pooled images	57,622	—
After duplicate/conflict exclusion	55,412	−2,210
After invalid-sample removal	55,409	−3

A conditional generative adversarial network (cGAN) was used to generate 150 labelled images each for Bird, Helicopter and UAV, or 450 synthetic samples in total. These samples were stored separately. The following "after" counts are an arithmetic sum of canonical training annotations and reported synthetic labels; they do not establish a physically merged, balanced training dataset.

Class	Original train boxes	Synthetic labels	Arithmetic after total	Change
Airplane	11,612	+0	11,612	0.00%
Bird	11,846	+150	11,996	1.27%
Helicopter	11,445	+150	11,595	1.31%
Kite	6,650	+0	6,650	0.00%
Parachute	6,650	+0	6,650	0.00%
UAV (Drone)	12,167	+150	12,317	1.23%
Total	**60,370**	**+450**	**60,820**	**0.75%**

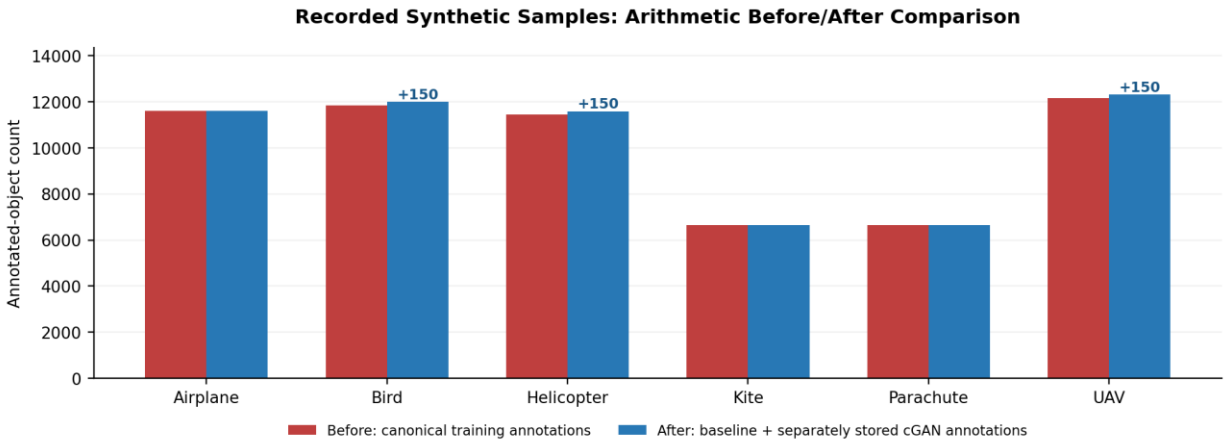


Figure 8. Comparison of original training annotations (red) and the arithmetic total after separately generated cGAN labels (blue).

The synthetic augmentation is targeted toward drone-related discrimination, not comprehensive class-frequency balancing. Kite and Parachute remain the minority classes. Accordingly, a genuinely balanced six-class dataset and its class-distribution figure are not available; no such result is claimed here.

4. Methodology

The methodology consists of dataset consolidation and verification, class-aware preprocessing, partition preparation, targeted augmentation, model-module definition and application-level detection processing. The numbered steps below distinguish completed data-processing work from architecture components whose full six-class comparative training has not been established.

Step 1. Dataset acquisition and loading. AOD-4 and MBAOD image collections and their corresponding annotations are identified and read. The original class definitions and the image–label correspondence are retained to support unification into a single dataset.

Step 2. Image validation and integrity screening. Image/annotation pairs are inspected for readability, class consistency and malformed data. The audit identifies three invalid samples, which were removed from the consolidated set.

Step 3. Exact-duplicate removal. Source images are compared using SHA-256 hashes to identify repeated or conflicting copies. The recorded cleaning procedure excludes 2,210 images, leaving 55,412 before invalid-sample removal.

Step 4. Near-duplicate grouping. Perceptual hashing (pHash) and structural similarity (SSIM) are used to identify visually related frames. Similarity grouping supports splitting decisions intended to reduce near-duplicate leakage between partitions.

Step 5. Class-label encoding and annotation correction. AOD-4 identifiers are remapped to the unified six-class order: 0 Airplane, 1 Bird, 2 Helicopter, 3 Kite, 4 Parachute, 5 UAV. Normalized YOLO bounding boxes are corrected where necessary; the audit records 55 clipped boxes.

Step 6. Clean dataset preparation and splitting. The canonical audit identifies 55,409 retained images allocated into 38,786 training, 8,311 validation and 8,312 testing images. A separate active research subset contains 20,620 training entries; the two dataset configurations are not treated as identical.

Step 7. Stratified K-fold partitioning. The 20,620-entry active training pool is assigned to ten stratified validation folds. Each fold contains 2,062 validation and 18,558 training entries. Fold files exist, but model training and accuracy measurements for those folds have not been completed.

Step 8. Training-region crop extraction. Bird, Helicopter and UAV target regions are isolated using their annotations. The recorded cGAN experiment prepares 300 crops per selected class (900 crops in total), resized to 64 × 64 pixels.

Step 9. Conditional GAN augmentation. The conditional generator takes a latent vector together with class information and produces class-specific target samples. The audit records 450 labelled synthetic composites, 150 per selected class. Their storage remains separate from the canonical training images.

Step 10. YOLO baseline and hybrid model definitions. A YOLO-based localisation component and five hybrid architecture modules are represented in the source code: CNN, LSTM, Transformer, BiLSTM-Attention and cGAN-Transformer extensions. The available baseline checkpoint is YOLOv8n; completed six-class hybrid training results are not established.

Step 11. Image enhancement and post-processing. The application pipeline incorporates contrast-limited adaptive histogram equalisation (CLAHE), morphological top-hat processing and non-maximum suppression (NMS) to improve visualisation and reduce overlapping detection boxes. Tracking and alerting components support the surveillance interface.

Step 12. Output and evaluation framework. The implemented components support localisation and class-aware visual output. Formal comparative model evaluation would use measured precision, recall, F1-score and detection mAP on compatible test partitions; these metrics are not presented as completed results in this methodology.

4.1 Methodology flowchart

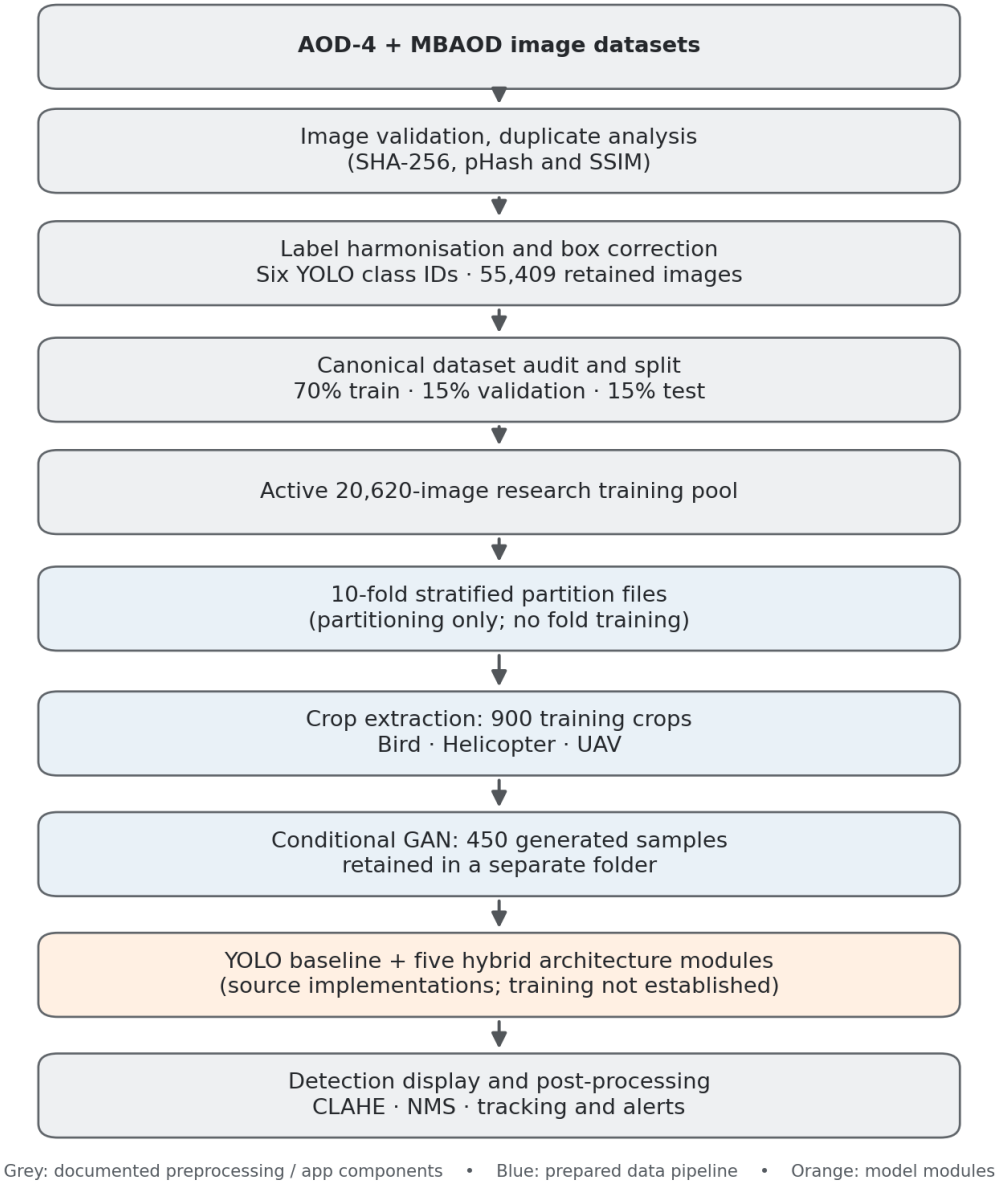


Figure 9. Methodology flowchart showing dataset preparation, augmentation and research implementation components.